

Deep Learning for Solving Dynamic Models

Session 4: Overlapping generations with DEQNs

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Based on Azinovic, Gaegauf & Scheidegger (2022, *IER* 63(4), 1471–1525) and Krueger & Kübler (2004, *JEDC*).

Roadmap

Part I: What is an OLG model?

Why overlapping generations; the cohort structure; the DEQN recipe in one line.

Part I½: The simplest OLG, two periods (Diamond, 1965)

One decision, one Euler equation, one closed form. Notebook 04_00.

Part II: The analytic six-agent model (Krueger & Kübler, 2004)

Euler equation and closed form derived; recursive equilibrium; cost function; network; validation. Notebook 04_01.

Part III: The 56-cohort benchmark of Azinovic, Gaegauf & Scheidegger (2022)

CRRA, hump-shaped labor, two assets, two constraints, persistent shocks. Slides only: the working code, with the paper's trained weights, is the `DeepEquilibriumNets` repository.

Learning objectives

By the end of this lecture you should be able to:

- ▶ Derive the Euler equation and the closed-form savings rate of a log-utility OLG economy, and say precisely which two assumptions make the closed form possible
- ▶ Construct a finite-dimensional overlapping-generations equilibrium with cohort-specific savings and lifecycle heterogeneity
- ▶ Derive equilibrium conditions and build the mean-squared Euler-residual loss for a multi-cohort OLG model
- ▶ Implement the DEQN recipe: model specification, equilibrium definition, cost function, network training, and validation
- ▶ Validate a learned solution against the closed-form equilibrium in the analytic Krueger–Kuebler model
- ▶ Read a research-scale benchmark (56 cohorts, two assets, two constraints, adjustment costs) as the same recipe with a larger G , and recognise when its optimisation, not its economics, is what fails

Why Overlapping Generations?

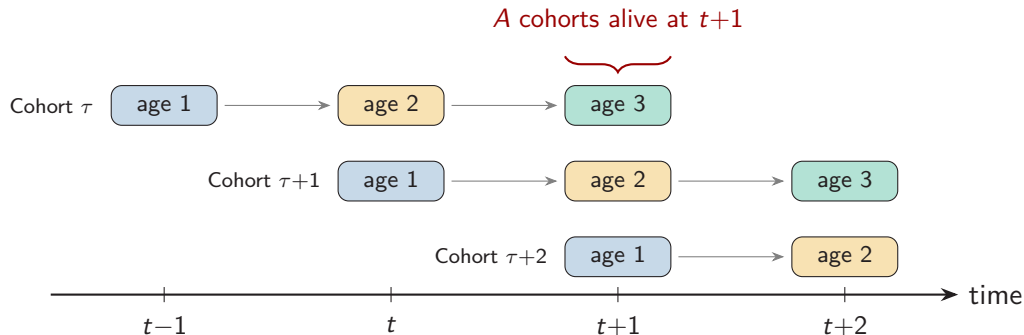
The economic content

- ▶ **Lifecycle:** agents are born, work, save, retire, die. A representative infinitely-lived household cannot speak to pensions, retirement, inheritance, or cohort-specific shocks.
- ▶ **Demographics:** at any t , many cohorts coexist, each with its own asset holdings and labor income.
- ▶ **Finite asset distribution:** with A cohorts, the cross-sectional wealth distribution is an A -vector, not an infinite-dimensional object. **Perfect for DEQNs:** state stays finite-dimensional with realistic heterogeneity.

Two canonical versions we solve today

1. $A = 6$ analytic model (Krueger & Kübler, 2004): closed form $\Rightarrow$ ground-truth validation.
2. $A = 56$ leading-example benchmark (Azinovic, Gaegauf & Scheidegger, 2022, §3): research-scale, no closed form.

OLG Structure: Overlapping Cohorts



At each period t we have A coexisting cohorts. A newborn enters (age 1); the oldest dies (age A).

Key observation: the number of types is finite (A), so the cross-sectional state is finite-dimensional.

The DEQN Recipe in One Line

From model to neural network (to be applied twice in today's lecture)

1. Write down the **economic model**: preferences, technology, constraints, shocks.
2. Define the **equilibrium**: choices + prices that satisfy optimization + market clearing.
3. Write the **equilibrium conditions** as a system $G(x, f(x)) = 0$.
4. Replace f by a **neural network** $\mathcal{N}_\theta$ and form the **cost function**
$$\mathcal{L}(\theta) = \frac{1}{|D|} \sum_j \|G(x_j, \mathcal{N}_\theta(x_j))\|^2.$$
5. Sample training states D , minimize $\mathcal{L}(\theta)$ by SGD.

The only thing that depends on the model

The **equilibrium conditions** G . Everything else, the network, the SGD loop, the sampling method, is generic. **So to solve a new model, write down its G .**

Today's deliverable: derive G explicitly for both the analytic and the benchmark OLG models.

Diamond (1965): Two Periods, One Decision

Households. Live two periods.

- ▶ **Young:** work (one unit), earn w_t ; consume c_t^Y , save s_t in capital: $c_t^Y + s_t = w_t$.
- ▶ **Old:** no labor; sell the capital, consume the proceeds: $c_{t+1}^O = r_{t+1} s_t$.
- ▶ Preferences $\ln c_t^Y + \beta \mathbb{E}_t \ln c_{t+1}^O$, $\beta = 0.7$.

Firm. Same technology as later:

$\eta(z)K^\alpha L^{1-\alpha} + (1 - \delta(z))K$, $L = 1$. The old hold all the capital: $K_t = s_{t-1}$.

$$r_t = \alpha \eta(z_t) K_t^{\alpha-1} + 1 - \delta(z_t), \quad w_t = (1 - \alpha) \eta(z_t) K_t^\alpha.$$

State

(z_t, k_t) with $k_t = K_t$ the capital the old carry in. Prices are functions of it.

Shocks

The same four i.i.d. states (η, δ) as in Part II, probability $\frac{1}{4}$ each.

Why start here. Session 2's Brock–Mirman model also had one Euler equation and a closed form. What is new is that *two different households are alive at the same date*, so the cross-section is part of the state. Two cohorts is the smallest structure with that property.

The Euler Equation, Derived and Solved

Substitute both budget constraints: the young choose s to maximise

$$\ln(w_t - s) + \beta \mathbb{E}_t \ln(r_{t+1} s).$$

Step 1. $\ln(r_{t+1}s) = \ln r_{t+1} + \ln s$: the return and the choice **separate**.

Step 2. First-order condition:

$$\frac{1}{w_t - s} = \beta \mathbb{E}_t \left[\frac{r_{t+1}}{r_{t+1} s} \right] = \frac{\beta}{s} \quad \Rightarrow \quad s_t = \frac{\beta}{1 + \beta} w_t = 0.4118 w_t$$

Two observations

1. The return cancels. Under log utility the savings *rate* does not depend on the distribution of r_{t+1} , whether it is risky or safe, persistent or i.i.d.
2. The old have no wage, so today's wage is the household's entire lifetime resource. There is nothing to borrow against.

The DEQN for It

Unknown: one function $s = f(z, k)$.

Network: $\mathcal{N}_\theta(z, k) \rightarrow \hat{\sigma} \in (0, 1)$, a savings fraction of the wage, $\hat{s} = \hat{\sigma} w(z, k)$. So $0 \leq \hat{s} \leq w$ and $c^y = w - \hat{s} > 0$ by construction.

Residual (relative Euler error): tomorrow's capital is $K' = \hat{s}$, tomorrow's return $r'(z', K')$ follows for each of the four shocks, the old consume $c^{o'} = r'\hat{s}$, and

$$e(z, k) = \frac{(\beta \mathbb{E}[r'/c^{o'}])^{-1}}{w - \hat{s}} - 1.$$

Loss: mean of e^2 over a cloud of states that is re-simulated under the current policy after every gradient step.

What is still missing

Tomorrow's policy does not enter: the old simply consume what they hold, so the residual calls $\mathcal{N}_\theta$ *once*. With three or more cohorts, cohort h 's Euler equation needs the consumption of a cohort $h + 1$ that does not exist yet, so the residual must evaluate $\mathcal{N}_\theta$ at the next state too. That self-reference is what a Deep Equilibrium Net handles, and we return to it on II.7a.

The notebook computes r' and $c^{o'}$ explicitly although they cancel, so that the code carries over to the six-cohort case unchanged.

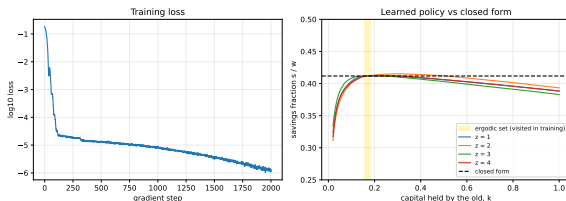
Finger Exercise: Two Periods

1. Part II will state a general formula for the savings rate of a cohort with $A - h$ periods still ahead: $\beta_h = \beta (1 - \beta^{A-h}) / (1 - \beta^{A-h+1})$. Evaluate it at $A = 2$, $h = 1$.
2. Suppose tomorrow's return r_{t+1} becomes *riskier* (same mean, more spread). What happens to s_t ? Why?
3. Now suppose the old *also* received a wage w_{t+1} . Would the young still save a fixed fraction of w_t ? Would they ever want to borrow?

3 minutes. Then open `04_00_OLG_Diamond_Warmup.ipynb`: about one minute on a laptop. The network should report $\hat{\sigma} = 0.4118$ at every state the economy visits, and drift away from it *outside* that set, because the loss is only ever evaluated where the policy goes (Session 2, "Residual training only tests states the policy visits"). That is why training uses a simulated cloud.

Solution: Two Periods

1. $\beta_1 = \beta(1 - \beta)/(1 - \beta^2) = \beta/(1 + \beta)$: the two-period result is the $A = 2$ case of the general formula.
2. Nothing changes. $\mathbb{E} \ln(r's) = \mathbb{E} \ln r' + \ln s$, so the spread of r' only shifts a constant that the first-order condition does not see. (With CRRA, $\gamma > 1$, the young would save more, a precautionary motive, and the closed form is lost.)
3. No, and yes. A future wage is a second lifetime resource; the young would want to borrow against it when w_t is low, and saving would depend on (w_t, w_{t+1}, r_{t+1}) jointly. This is the 56-cohort benchmark of Part III in miniature.



II.1 Setup: the Analytic Model

- ▶ Time discrete: $t = 0, 1, 2, \dots, \infty$.
- ▶ Agents live for $A = 6$ periods.
- ▶ **One representative household per cohort**; a new one is born every period.
- ▶ **No uncertainty about lifetime**: all agents live exactly A periods.
- ▶ Exogenous aggregate shock $z_t \in \{1, 2, 3, 4\}$ drawn **i.i.d.** with transition $\pi_{ss'} = 0.25$.
- ▶ **Log utility** $u(c) = \ln c$, chosen because it gives a **closed-form** equilibrium (Krueger & Kübler, 2004).
- ▶ **Labor**: only the youngest cohort works: $\ell^1 = 1$, $\ell^h = 0$ for $h \geq 2$.

Shocks: $z_t = (\eta_t, \delta_t)$

z	$\eta(z)$	$\delta(z)$	π
1	0.95	0.5	0.25
2	1.05	0.5	0.25
3	0.95	0.9	0.25
4	1.05	0.9	0.25

Calibration

$\alpha = 0.3$, $\beta = 0.7$, $\gamma = 1$ (log); $\underline{a} = 0$.

II.2 Households: Preferences and the Savings Technology

Agents supply their labor **inelastically** for a market wage w_t .

An agent **currently of age h** alive at t maximises remaining time-separable discounted expected lifetime utility ($\beta < 1$):

$$\mathbb{E}_t \left[\sum_{i=0}^{A-h} \beta^i \ln(c_{t+i}^{h+i}) \right] \quad (1)$$

Households can save a unit of consumption to obtain a unit of capital good next period, denoted a_t^h .

The savings become **capital in the next period**:

$$a_t^h = k_{t+1}^{h+1}, \quad \forall t, \forall h \in \{1, \dots, A-1\}. \quad (2)$$

II.3 Households: Constraints, Budget, Boundary

Households **cannot die with debt**. Borrowing is bounded:

$$a_t^h \geq \underline{a}. \quad (3)$$

In the analytic model we set $\underline{a} = 0$: **no borrowing** for any cohort.

At time t , households sell their capital to the firm at market price $r_t > 0$. The per-period **budget constraint** of cohort h reads

$$c_t^h + a_t^h = r_t k_t^h + \ell^h w_t. \quad (4)$$

Boundary conditions. Newborns enter with no inheritance, and the oldest cohort dies without saving anything:

$$k_t^1 = 0, \quad a_t^A = 0.$$

II.4 Firms and Markets

There is a single representative firm with **Cobb–Douglas** production.

TFP η and depreciation δ depend on the exogenous shock z alone (see II.1 table).

Each period, **after the shock has realized**, the firm buys capital and hires labor to maximize profits, taking prices as given.

The stochastic production function is

$$f(K, L, z) = \eta(z) K^\alpha L^{1-\alpha} + K(1 - \delta(z)).$$

There are **competitive spot markets** for the consumption good, capital, and labor.

Firm FOCs (closed form, from Cobb–Douglas):

$$w_t = (1 - \alpha) \eta(z_t) K_t^\alpha L_t^{-\alpha}, \quad r_t = \alpha \eta(z_t) K_t^{\alpha-1} L_t^{1-\alpha} + (1 - \delta(z_t)).$$

In the analytic model $L_t = \sum_{h=1}^A \ell^h = 1$ (only age-1 works), so the firm problem reduces to $\max_{K \geq 0} \eta(z) K^\alpha - (r_t - (1 - \delta(z)))K$. **Prices are a function of the state alone**, which is why the network never has to learn them here.

II.5 Competitive Equilibrium: Definition

Definition 1 (competitive equilibrium)

Given initial conditions z_0 , $\{k_0^h\}_{h=1}^{A-1}$, a competitive equilibrium is a collection of choices for households $\{(c_t^h, a_t^h)_{h=1}^A\}_{t=0}^\infty$, for the representative firm $(K_t, L_t)_{t=0}^\infty$, and prices $(r_t, w_t)_{t=0}^\infty$, such that:

1. Given $(r_t, w_t)_{t \geq 0}$, household choices maximize the lifetime utility (1) subject to the savings-to-capital identity (2), the borrowing limit (3), and the per-period budget (4):

$$a_t^h = k_{t+1}^h; \quad a_t^h \geq \underline{a}; \quad c_t^h + a_t^h = r_t k_t^h + \ell^h w_t.$$

2. Given (r_t, w_t) , the firm maximises profits:

$$(K_t, L_t) \in \arg \max_{K, L \geq 0} f(K, L, z_t) - r_t K - w_t L.$$

3. **All markets clear:** for all t , $L_t = \sum_{h=1}^A \ell^h$, $K_t = \sum_{h=1}^A k_t^h$.

II.6 Household Optimality: Deriving the Euler Equation

Cohort h at date t chooses $\{c, a\}$ for its remaining $A - h + 1$ periods. Attach ν to the budget constraint (4) and $\lambda \geq 0$ to the borrowing limit (3):

$$\mathcal{L} = \mathbb{E}_t \sum_{i=0}^{A-h} \beta^i \left[\ln c_{t+i}^{h+i} + \nu_{t+i}^{h+i} (r_{t+i} k_{t+i}^{h+i} + \ell^{h+i} w_{t+i} - c_{t+i}^{h+i} - a_{t+i}^{h+i}) + \lambda_{t+i}^{h+i} a_{t+i}^{h+i} \right], \quad k_{t+i+1}^{h+i+1} = a_{t+i}^{h+i}.$$

- FOC in c_t^h : $\nu_t^h = 1/c_t^h$, the marginal utility of cash.
- FOC in a_t^h : $-\nu_t^h + \lambda_t^h + \beta \mathbb{E}_t [\nu_{t+1}^{h+1} r_{t+1}] = 0$, since a_t^h becomes k_{t+1}^{h+1} and earns r_{t+1} .
- Complementary slackness: $\lambda_t^h \geq 0$, $a_t^h \geq 0$, $\lambda_t^h a_t^h = 0$.

$$\boxed{\frac{1}{c_t^h} = \beta \mathbb{E}_t \left[\frac{r_{t+1}}{c_{t+1}^{h+1}} \right] + \lambda_t^h} \quad \text{terminal cohort: } a_t^A = 0, \quad c_t^A = r_t k_t^A.$$

Here $\lambda^h \equiv 0$. No cohort has future labor income (only the newborn works, once), so there is nothing to borrow against and $a_t^h \geq 0$ never binds: what remains is the **interior** Euler equation. Part III keeps λ and adds a second multiplier, because a hump-shaped wage profile gives the young a reason to borrow.

II.6a Why This Model Has a Closed Form, and Why the Benchmark Does Not

Two assumptions

1. **Log utility.** $\mathbb{E} \ln(r' a) = \mathbb{E} \ln r' + \ln a$, so the savings rate does not depend on the return distribution: income and substitution effects of a return shock cancel. Persistent shocks would change nothing here; i.i.d. shocks only keep the aggregate dynamics simple.
2. **No future labor income.** Only the newborn works, once. Cash-on-hand $W_t^h = r_t k_t^h + \ell^h w_t$ is the household's entire state, there is nothing to borrow against, and $\lambda \equiv 0$.

What the 56-cohort benchmark drops

- ▶ CRRA with $\gamma = 2$: the return no longer separates, and a riskier r' raises saving.
- ▶ Hump-shaped ℓ^h : the young have wages ahead of them and want to borrow, so the constraint binds, and so does a collateral constraint on bonds.

No closed form; multipliers in the network; a numerical solver is needed.

II.6b Backward Induction to β_h

Write W for cash-on-hand and $V_h(W)$ for the value of a cohort of age h .

- **Age A :** no tomorrow. $c = W$, $V_A(W) = \ln W$.
- **Age $A - 1$:** $\max_a \ln(W - a) + \beta \mathbb{E} \ln(r' a)$. FOC $\frac{1}{W-a} = \frac{\beta}{a} \Rightarrow a = \frac{\beta}{1+\beta} W$,
 $V_{A-1}(W) = (1 + \beta) \ln W + \text{const.}$
- **Age $A - 2$:** $\max_a \ln(W - a) + \beta(1 + \beta) \mathbb{E} \ln(r' a) \Rightarrow a = \frac{\beta + \beta^2}{1 + \beta + \beta^2} W$,
 $V_{A-2}(W) = (1 + \beta + \beta^2) \ln W + \text{const.}$
- **Age h , by induction:**

$$\beta_h = \frac{a}{W} = \frac{\beta + \beta^2 + \dots + \beta^{A-h}}{1 + \beta + \dots + \beta^{A-h}} = \beta \frac{1 - \beta^{A-h}}{1 - \beta^{A-h+1}}, \quad h = 1, \dots, A - 1.$$

The constant absorbs $\mathbb{E} \ln r'$ at every step, which is why the return distribution never reaches the decision. W is the inc_t^h of the budget constraint (4), so the sigmoid savings-fraction head of II.9 has the same shape as the closed form.

II.7 The DEQN Cost Function

Explicit cost function

$$\mathcal{L}_{D_{\text{train}}}(\theta) := \frac{1}{|D_{\text{train}}|} \frac{1}{A-1} \sum_{x_j \in D_{\text{train}}} \sum_{i=1}^{A-1} (e_{\text{REE}}^i(x_j))^2$$

Relative Euler-equation residual (cohort i):

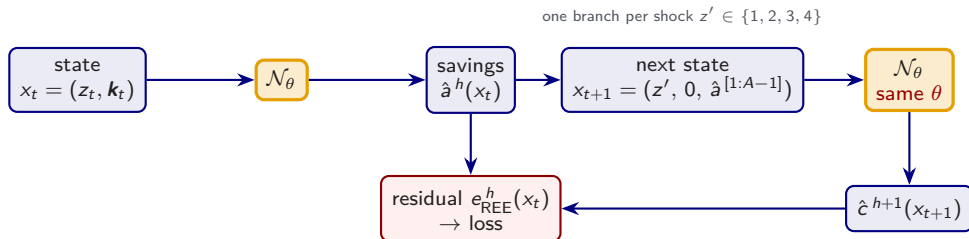
$$e_{\text{REE}}^i(x_j) := \frac{\left(u'\right)^{-1}\left(\beta \mathbb{E}\left[r\left(\hat{x}_{j,+}\right) u'\left(\hat{c}^{i+1}\left(\hat{x}_{j,+}\right)\right)\right]\right)}{\hat{c}^i\left(x_j\right)} - 1.$$

Next state and expectation. $\hat{x}_{j,+} = (z_+, 0, \hat{a}^{[1:A-1]}(x_j))$, the newborn entering with $k^1 = 0$; $\mathbb{E}$ is the exact four-term sum over z_+ .

No market-clearing residual. Next period's capital is defined as $K_{t+1} = \sum_h \hat{a}^h(x_j)$, so the capital market clears by construction. Part III needs one for bonds.

In the code (04_01) the loss carries two feasibility penalties, on negative consumption and negative aggregate capital, weight 10; they keep early training out of the region where $\ln c$ is undefined and are inactive thereafter.

II.7a Where the Network Appears Twice



Cohort h 's Euler equation needs c_{t+1}^{h+1} , the consumption of a household *one year older* next period. Nobody hands that to us, so the network supplies it: one call at x_t , one call at each successor state. **Five forward passes per residual**, all with the same θ .

Why this matters

The loss is a **fixed-point condition on θ** , not a regression: there is no target to fit, only the requirement that the network be consistent with itself one period ahead. Gradients flow through *both* calls. This is also why there is no contraction theorem and the stopping rule has to be empirical (Session 2, Algorithm 1).

II.8 From a Sequence to a Function (FREE)

Definition 1 describes an equilibrium as *sequences* $\{c_t^h, a_t^h, r_t, w_t\}_{t \geq 0}$. A network represents a *function*. The link is a sufficient state.

$(z_t, \mathbf{k}_t)$ is all a household needs to know

- *Today*: $K_t = \sum_h k_t^h$ and z_t pin down (r_t, w_t) , hence every cohort's cash-on-hand.
- *Tomorrow*: if behaviour is the same in the same state, $(z_{t+1}, \mathbf{k}_{t+1})$ depends only on $(z_t, \mathbf{k}_t)$, the shock, and that behaviour.

So we look for a **recursive** equilibrium: time-invariant policies with $\mathbf{k}' = (0, a^1(z, \mathbf{k}), \dots, a^{A-1}(z, \mathbf{k}))$.

Functional rational expectations equilibrium (FREE)

$$f : \{1, 2, 3, 4\} \times \mathbb{R}^A \rightarrow \mathbb{R}^{A-1}, \quad f(z_t, \mathbf{k}_t) = [a_t^1, \dots, a_t^{A-1}]^\top, \quad \beta \mathbb{E} \left[\frac{r_{t+1} u'(c_{t+1}^{h+1})}{u'(c_t^h)} \right] - 1 = 0 \quad \forall h,$$

i.e. $G([z_t, \mathbf{k}_t], f([z_t, \mathbf{k}_t])) = \mathbf{0}$ at every state.

Existence and stationarity are *assumed*; in Part II the closed form is such a function, so here the assumption is verified. f is the unknown the equilibrium defines; $\mathcal{N}_\theta$ next slide is the parametric family we search in.

II.9 Deep Equilibrium Net: Architecture (Analytic)

$$\mathcal{N}_\theta : \{1, 2, 3, 4\} \times \mathbb{R}^A \rightarrow \mathbb{R}^{A-1}$$

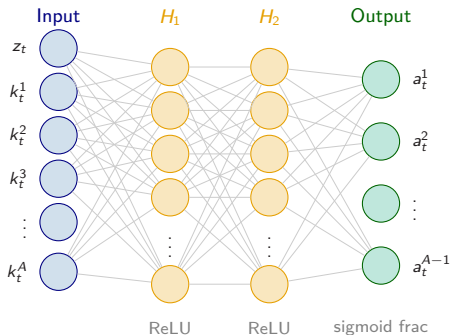
$$\mathcal{N}_\theta\left(\begin{bmatrix} z_t \\ \mathbf{k}_t \end{bmatrix}\right) = \begin{bmatrix} a_t^1 \\ a_t^2 \\ \vdots \\ a_t^{A-1} \end{bmatrix}$$

such that

$$G\left(\begin{bmatrix} z_t \\ \mathbf{k}_t \end{bmatrix}, \mathcal{N}_\theta\left(\begin{bmatrix} z_t \\ \mathbf{k}_t \end{bmatrix}\right)\right) \approx \mathbf{0}.$$

Sigmoid savings-fraction output $\Rightarrow$

$0 \leq a_t^h \leq \text{inc}_t^h$ **by construction**, so $a_t^h \geq 0$
(borrowing constraint) and $c_t^h \geq 0$ both hold
at every iteration without an explicit multiplier.



2 hidden layers $\times$ (100, 50) units; single forward pass.

Companion Notebook Hand-over

Open notebook `04_01_OLG_Analytic_DEQN_persistent.ipynb` (the persistent-simulation primary; the exogenous-sampling companion is `04_04_OLG_Analytic_DEQN_exogenous.ipynb`; if you skipped it, `04_00` is the two-period warm-up with the same structure in sixty lines). It implements the model, equilibrium conditions, loss, FREE map, architecture, training, and validation covered in Part II.

Cell-to-slide map

Deck slide	Notebook section
II.1–II.6 Model and equilibrium	§1–3 parameters, state, network/policy
II.7–II.9 Loss, FREE, architecture	§4 <code>compute_cost()</code> + training-data construction
II.10–II.13 Training/validation	§5–8 mini-batches, diagnostics, policy stability, closed-form checks

II.10 Sampling and the Training Loop

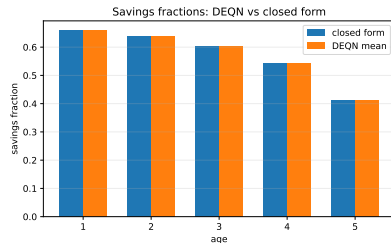
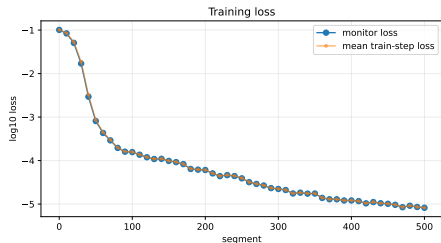
Cloud method (Azinovic, Gaegauf & Scheidegger, 2022)

1. **Initialisation.** Sample a cloud of initial states $D_0 = \{x_j^0\}_{j=1}^N$; initialise θ^0 .
2. **Episode k .**
 - (a) **Simulate.** Draw fresh shocks $\{z_{+,j}\}$ from π ; roll the cloud forward one period using the **current** policy $\mathcal{N}_{\theta^k}$: $x_j^{k+1} = (z_{+,j}, 0, \mathcal{N}_{\theta^k}^{[1:A-1]}(x_j^k))$.
 - (b) **Mini-batch SGD.** Take steps Adam steps on $\mathcal{L}_{D_k}(\theta)$ using mini-batches from D_k .
3. **Stop** when $\log_{10} \mathcal{L}$ falls below the target.

Definitions. **Episode** = one simulate-then-train pass. **Epoch** = one pass through the full cloud.

Sampling from the ergodic set of the current policy makes the cost function focus on states the economy actually visits.

II.11 Training Results: Analytic Model (teaching-mode run)



Reading the plots

Left: monitor loss and mean train-step loss per segment, $\log_{10}$ scale; the cloud is re-simulated each segment.

Right: mean savings fraction by cohort against the closed form β_h ; the DEQN matches it to 1.4×10^{-4} .

From 04_01, RUN_MODE = "teaching": about 45 minutes on a laptop CPU, so the stored outputs are that run; use "smoke" (~ 30 s) in class.

Finger Exercise: Closed-Form Savings Rates

Task (3 minutes). Take β_h as derived on II.6b,

$$\beta_h = \frac{a_t^h}{\text{inc}_t^h} = \beta \frac{1 - \beta^{A-h}}{1 - \beta^{A-h+1}}, \quad \text{inc}_t^h = r_t k_t^h + \ell^h w_t, \quad h = 1, \dots, A-1,$$

and evaluate it at $\beta = 0.7$, $A = 6$: compute $\beta_1, \beta_2, \dots, \beta_5$.

Then interpret.

- ▶ What happens to the savings *rate* as a cohort approaches retirement, and why?
- ▶ Why do younger cohorts save more as a rate, even though they are poorer?
- ▶ These five numbers are the **validation target** of Krueger & Kübler (2004): the DEQN has to reproduce them without ever being told what they are.

Solution: Closed-Form Savings Rates

$$\beta_h = 0.7 \cdot \frac{1 - 0.7^{6-h}}{1 - 0.7^{7-h}}$$

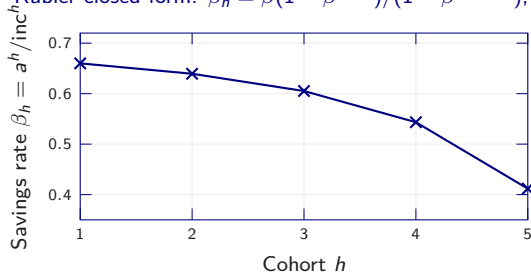
Cohort h	Periods remaining ($A - h$)	Savings rate β_h
1	5	$0.7 \cdot (1 - 0.7^5)/(1 - 0.7^6) \approx 0.660$
2	4	$0.7 \cdot (1 - 0.7^4)/(1 - 0.7^5) \approx 0.639$
3	3	$0.7 \cdot (1 - 0.7^3)/(1 - 0.7^4) \approx 0.605$
4	2	$0.7 \cdot (1 - 0.7^2)/(1 - 0.7^3) \approx 0.543$
5	1	$0.7 \cdot (1 - 0.7)/(1 - 0.7^2) \approx 0.412$
6	0	0 (consumes everything)

Interpretation

- ▶ Savings rate **monotonically declines** with age: as the horizon shrinks, saving less is optimal.
- ▶ Youngest cohort saves $\sim 66\%$ of its income, longest horizon to enjoy the returns.
- ▶ At A : terminal boundary $a^A = 0$, consume everything.

II.13 DEQN vs. Analytical Solution

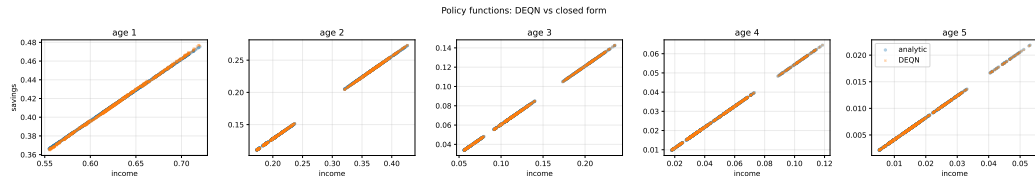
Krueger-Kübler closed form: $\beta_h = \beta(1 - \beta^{A-h})/(1 - \beta^{A-h+1})$, $\beta = 0.7$, $A = 6$



Validation strategy

The points are the exact closed form. The DEQN should reproduce them cohort by cohort; the next slide checks it state by state. Production target in AGS (2022): mean relative Euler error below 10^{-3} .

II.13' Policy Functions State by State



One panel per cohort: savings a^h against cash-on-hand inc^h on the simulated ergodic cloud. The closed form is a line through the origin with slope β_h ; the DEQN should lie on it at every income level.

The ergodic gross return is about 0.69, below one because $\delta \in \{0.5, 0.9\}$, and the young still save two thirds of their wage: capital is the only store of value in this economy.

II. Take-away and Hands-on

What we just did

1. OLG model with 6 cohorts, log utility, 4 i.i.d. shocks (II.1–II.4).
2. Equilibrium definition + conditions (II.5–II.6).
3. Conditions $\rightarrow$ **cost function** (II.7).
4. Equilibrium map $f \rightarrow$ neural network $\mathcal{N}_\theta$ (II.8–II.9).
5. Trained by sampling + SGD (II.10–II.11), **validated** against the closed form (II.13).

Hands-on

Open 04_01_OLG_Analytic_DEQN_persistent.ipynb. Run `RUN_MODE = "smoke"` (~ 30 s) now; the stored outputs are already the converged "teaching" run (~ 45 min on a laptop CPU), and "production" chases AGS Table 3 accuracy. Both notebooks expose a `SAMPLING_MODE = "simulation" / "exogenous"` switch (the persistent primary uses "simulation", the companion 04_04_OLG_Analytic_DEQN_exogenous.ipynb uses "exogenous"); the model and loss are identical, only the training cloud differs.

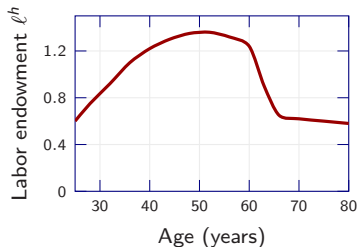
III.1 What Changes at Research Scale: AGS (2022), §3

Same recipe as Part II. Seven things change:

- ▶ $A = 56$ periods (ages 25–80), one period = one year.
- ▶ CRRA $u(c) = c^{1-\gamma}/(1-\gamma)$, $\gamma = 2$, **not** log.
- ▶ **Hump-shaped** labor ℓ^h (right), so the young have wages ahead of them.
- ▶ **Persistent** Markov shocks on (η, δ) , not i.i.d.
- ▶ **Two assets**: capital k^h and a one-period bond b^h in zero net supply, $\sum_h b_t^h = 0$, traded at an endogenous price p_t .
- ▶ **Two constraints**: $k'^h \geq 0$ and $k'^h + \kappa b'^h \geq 0$.
- ▶ Capital adjustment costs $\Psi^h = \frac{\zeta}{2}(k'^h - rk^h)^2$.

Budget: $c_t^h + k_{t+1}^{h+1} + p_t b_{t+1}^{h+1} + \Psi_t^h = r_t k_t^h + b_t^h + \ell^h w_t$. Born and die with nothing. Three markets clear: labor, capital, **and bonds**.

Stylised lifecycle labor (schematic)



This is the model that breaks the closed form.

CRRA kills the separation of II.6a; future wages give the young a reason to borrow, so the constraints bind and multipliers are back.

III.2 Equilibrium Conditions, and Where Each Multiplier Lands

Same Lagrangian as II.6, with $\lambda_b \geq 0$ on $k' \geq 0$ and $\mu \geq 0$ on $k' + \kappa b' \geq 0$. Write $\mathcal{A}_t^h := 1 + \zeta(k_t^{h'} - r_t k_t^h)$ for the marginal adjustment-cost wedge.

$$\begin{aligned} \frac{\partial}{\partial k'} : \quad u'(c_t^h) \mathcal{A}_t^h &= \beta \mathbb{E}[u'(c_{t+1}^{h+1}) r_{t+1} \mathcal{A}_{t+1}^{h+1}] + \underbrace{1 \cdot \lambda_b^h}_{\partial k' / \partial k'} + \underbrace{1 \cdot \mu^h}_{\partial(k' + \kappa b') / \partial k'} \\ \frac{\partial}{\partial b'} : \quad u'(c_t^h) p_t &= \beta \mathbb{E}[u'(c_{t+1}^{h+1})] + \underbrace{\kappa \mu^h}_{\partial(k' + \kappa b') / \partial b'} \quad (\lambda_b \text{ absent: } b' \text{ is not in that constraint}) \end{aligned}$$

The general rule, worth remembering

A multiplier enters each first-order condition weighted by **its constraint's derivative in that instrument**. The collateral constraint contains both assets, so μ appears twice, scaled by κ in the bond equation; the borrowing constraint contains only capital, so λ_b appears once.

Two KKT triples per cohort: $\lambda_b^h \geq 0, k'^h \geq 0, \lambda_b^h k'^h = 0$;
 $\mu^h \geq 0, k'^h + \kappa b'^h \geq 0, \mu^h(k'^h + \kappa b'^h) = 0$.

With $\mu = 0$ the bond equation is $p = \beta \mathbb{E}[u'(c')]/u'(c)$, the price of a riskless claim; a binding collateral constraint adds the shadow value $\kappa \mu / u'(c)$ that a constrained cohort will pay on top.

III.3 The Same Loss, at Twenty Times the Size

Stack everything, exactly as in II.7

$$\mathcal{L}(\theta) = \frac{1}{|D|} \frac{1}{4(A-1)+1} \sum_{x_j \in D} \left[\underbrace{\sum_{h=1}^{A-1} \left(e_{\text{REE},k}^h \right)^2 + \left(e_{\text{REE},b}^h \right)^2 + \underbrace{\left(\lambda_b^h k'^h \right)^2 + \left(\mu^h q^h \right)^2}_{\text{KKT, product form}}}_{\text{per cohort}} + \underbrace{\left(\sum_h \hat{b}'^h \right)^2}_{\text{bond market}} \right]$$

The network.

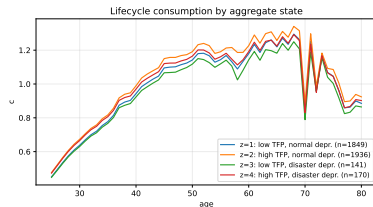
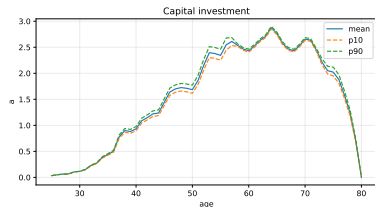
$$\mathcal{N}_\theta : \mathbb{R}^{240} \longrightarrow \mathbb{R}^{221}$$

- ▶ minimal state 113; input 240 after engineered aggregates and transition probabilities
- ▶ output $4 \times 55 + 1 = 221$: $\hat{k}'^h$, $\hat{\lambda}_b^h$, $\hat{b}'^h$, $\hat{\mu}^h$, and the **bond price** $\hat{p}_t$, an equilibrium price the network must learn
- ▶ two ReLU layers of 1000 units

Nothing conceptually new

Part II had $A - 1 = 5$ residuals per state. This has 221. The recipe, the cost function, the simulate-and-train loop, and the diagnostics are *identical*. What changes is only how hard the optimisation is, and that turns out to matter a great deal.

III.4 Solved: Lifecycle Profiles from the Paper's Network



Left: capital by age on the ergodic set, mean and 10/90 percentiles: the hump a representative-agent model cannot produce. **Right:** mean lifecycle consumption, one curve per aggregate state $z = (\eta, \delta)$.

Run it yourself, on the paper's trained weights

<https://github.com/sischei/DeepEquilibriumNets> → code/python-scripts/benchmark

- ▶ `python benchmark.py` loads the **final network weights of AGS (2022)** and regenerates the paper's plots in seconds, no training needed.
- ▶ `--train_from_scratch` reproduces the two-stage schedule: a GPU job, not a laptop one, which is the point of the next slide.

III.5 What a Failed Run Looks Like

Preparing this lecture, I tried to shrink the benchmark to a laptop-sized preset so you could train it in class. It does not work, and the way it fails is worth more than the run would have been. Four attempts, three diverged:

Diagnostic	healthy (Part II)	the runs that failed
$\log_{10} \mathcal{L}$	-5.1	+7 to +11
aggregate K across the cloud	a spread of states	runs off to 130, then a single point
share of the cloud with $c \leq 0$	0%	70–82%
policy drift	below tolerance	still moving after 400 segments

How to read it

- ▶ It looked *fine* for twenty segments, then K split away and never came back. A loss that is falling tells you nothing on its own.
- ▶ **Watch the state distribution, not the loss.** The simulated cloud is the training set; once it runs away, the residual is being minimised somewhere the economy never goes.
- ▶ Identical settings gave both a converged run and a diverged one: on CPU the arithmetic is not bit-reproducible, and this problem sits close enough to the edge for that to decide the outcome.

This is what Session 3's architecture search and loss balancing are for, and why AGS (2022) runs a two-stage schedule on a GPU.

Summary

What we accomplished

1. OLG models: a finite-dimensional family of heterogeneous-agent models, suited to DEQNs.
2. Applied the DEQN recipe **three times**:
 - ▶ **Part I½**: Diamond's two-period model, one Euler equation, solved by hand and by the network.
 - ▶ **Part II**: the 6-agent Krueger–Kübler (2004) analytic model, validated against the closed form ($\lesssim 10^{-3}$ mean Euler error).
 - ▶ **Part III**: the 56-cohort benchmark of AGS (2022) read off the slides: CRRA, hump-shaped labor, persistent shocks, two assets, two constraints, adjustment costs, and an honest look at what happens when its optimisation fails.
3. **Same recipe** every time: model, equilibrium conditions, cost, recursive equilibrium, network, training, validation.

One line to remember

Writing down a DEQN = writing down the equilibrium conditions **G**. Everything else (network, SGD, sampling) is boilerplate.

References and Further Reading

- ▶ Azinovic, M., Gaegauf, L., & Scheidegger, S. (2022). *Deep equilibrium nets*. International Economic Review 63(4), 1471–1525. [Part III; the reference implementation and the paper's trained weights are at <https://github.com/sischei/DeepEquilibriumNets>, code/python-scripts/benchmark].
- ▶ Krueger, D., & Kübler, F. (2004). *Computing equilibrium in OLG models with stochastic production*. Journal of Economic Dynamics & Control 28(7), 1411–1436. [04_01_OLG_Analytic_DEQN_persistent.ipynb implements their 6-agent analytic model].
- ▶ Diamond, P. A. (1965). *National debt in a neoclassical growth model*. American Economic Review 55(5), 1126–1150. [the two-period model of Part I½; 04_00_OLG_Diamond_Warmup.ipynb].
- ▶ Fischer, A. (1992). *A special Newton-type optimization method*. Optimization 24, 269–284. [FB-complementarity function, used in the **IRBC chapter (Ch. 3)**; mentioned in this lecture only as an alternative to the product-form KKT residuals adopted here.]
- ▶ Code: <https://github.com/sischei/DeepEquilibriumNets>.

Bridge: one more dimension to add

What we just did. Cohort structure changed the size of the residual system, one Euler equation per generation, derived once from the Lagrangian and stacked into one loss, but not the training principle. The two-period model gave the idea, the six-cohort model a closed form to validate against, and the 56-cohort benchmark added what breaks the closed form: CRRA, future wages, and the borrowing and collateral constraints the young actually hit, encoded as product-form KKT residuals.

Next (Session 5): the international real business cycle model. OLG stacked equations along the *age* dimension. IRBC stacks them along a *country* dimension instead: N Euler equations plus a world resource constraint on a $2N$ -dimensional state. Same loss, same training loop, and the point where a tensor-product grid is no longer an option.

Bring with you. The architecture search and loss balancing from Session 3: IRBC is the first model in this course whose residual components genuinely live on different scales.